MILITARY SPECIFICATION

MANUALS, TECHNICAL, CONVENTIONAL TO MICROFILM COMPATIBLE, CONVERSION OF

This specification has been approved by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification identifies and describes the general and specific requirements for conversion of conventional format maintenance instructions and illustrated parts breakdown technical manuals to microfilm compatible format technical manuals.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents, of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-38784 Manuals, Technical: General Style and Format Requirements

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, N.J. 08733 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC TMSS

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI Y14.2 Line Conventions and Lettering

(Application for copies should be addressed to the American National Standards Institute Incorporated, 1430 Broadway, New York, New York 10018).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies).

3. REQUIREMENTS

3.1 General. Maintenance instruction manuals (MIMs), illustrated parts breakdown (IPB) manuals, and combined MIM/IPB Manuals prepared to the general style and format requirements of MIL-M-38784 may not meet microfilm compatible standards as defined herein. Manuals that do not meet these standards shall be revised to make them compatible with microfilming in accordance with the requirements specified herein.

3.2 Data. Microfilm compatible technical manual data established by this specification shall be prepared in accordance with the data item description (DID) and the applicable technical manual contract requirement.

3.3 Microfilm Compatibility Criteria.

3.3.1 Type size. The type size shall have a minimum type face of 9 1/2 points and a maximum of 10 1/2 points in the deliverable products (manuscript, camera - ready copy, etc.). IPB Group Assembly Parts Lists (GAPL) pages shall be considered text. The type face size for text, dimensions, and call outs on illustrations, including schematics and diagrams, shall be a minimum type face of 8 points and a maximum type face of 9 1/2 points. The type face size for tables and tabular listings, including numerical and reference designation indexes, shall be a minimum of 8 points and a maximum of 9 1/2 points. Page numbers shall be 14 point bold face.

3.3.2 Artwork.

3.3.2.1 Photographs. Photographic halftone artwork shall not be included in technical manuals revised to this specification. Line tracings of photographs are acceptable. Prior to line tracing, existing photographs shall be reviewed for useability. When a photographic line tracing is prepared, proper definition of line work shall be used in lieu of photo retouching. The intended subject matter shall be highlighted and unnecessary background shall be eliminated. Items required for reference (location) shall be subdued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

3.3.2.2 Line drawings. Existing line drawings shall be reviewed to determine their useability. Line drawings with light, penciled, or narrow width lines shall be reworked using suitable materials that will produce products capable of maintaining consistent high-density tonal values.

3.3.2.2.1 Darkness and sharpness of lines. Line width shall be in accordance with ANSI Y14.2. Shading shall not be used. Patterns used to "code" fluid lines, mechanical linkage, electrical wiring, etc, or accentuate illustrations shall be clearly defined and suitable for microfilming. Patterns that tend to bleed lines together when reduced in size for microfilming are unacceptable.

3.3.2.3 Sidewise illustrations. Sidewise (turn page) illustrations shall be converted to direct viewing illustrations. After conversion of sidewise illustrations to direct viewing, the illustrations shall be easily readable, compatible with microfilming and meet the type size for text and callouts in 3.1.1. If conversion of an illustration on a single page reduces its legibility so that it does not meet the requirement contained herein, the illustration shall be redrawn as a multiple-sheet illustration or a foldout illustration.

3.3.2.4 Color. Color shall not be used in artwork. Line patterns (diagonals, dots, crosshatch, etc.) shall be used in place of color.

3.3.2.5 Overlays. The use of overleys for callouts on illustrations is not acceptable. Callouts shall be placed on the background area, close to, but free of the illustrated item.

3.3.2.6 Engineering drawings. The copy print of engineering drawings must conform to the type requirements for reproducible material and shall be legible for reproduction when reduced to technical manual size. If the type size and legibility requirements cannot be met with existing engineering drawings, those drawings must be redrawn and prepared as foldout illustrations in accordance with 3.3.7.

3.3.2.7 Foldout illustrations. (See figure 1) Existing foldout illustrations shall be checked to determine acceptability for microfilming. To be acceptable, an existing foldout illustration must be capable of being segmented in 7 to 7 1/2 inch wide increments of image area. There shall be no data (nomenclature, symbols, values, vertical lines) with the exception of horizontal lines placed closer than 1/8 inch from the image area width limit. Foldout illustrations which do not meet these criteria shall be redrawn. Connecting (Horizontal lines shall be extended to the edges of each page (frame) to facilitate alignment with adjacent increments.

3.3.2.7.1 Foldout-foldup illustrations. Foldout-foldup illustrations are not acceptable. Existing foldout-foldup illustrations must be redrawn as foldout illustrations.

3.3.2.7.2 Location of foldout illustrations. Foldout illustrations, like other illustrations, must be placed as near as possible to the related text. If the foldout illustrations are at the end of the manual, each foldout illustration must be moved to the location in the manual that the foldout illustration relates to. The relocated foldout illustrations must be appropriately numbered to correspond with their new locations in the manual.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

3.3.2.7.3 Foldout illustration increment numbering. Each foldout illustration increment shall include the technical manual number in the upper right corner of the increment. Each increment shall include the page number and change number, when applicable, of the foldout illustration in the lower right corner of the increment. The first increment of a foldout illustration shall include the figure title as follows:

"Figure No.____ Figure Title (Sheet 1 of ____)"

The remaining increments shall be as follows:

"Figure No.____ (Sheet ____)".

3.3.2.7.4 Classification markings - foldout illustrations. If a foldout illustration contains classified material, each increment of the illustration shall be marked at the top and bottom with the highest classification of the content of the foldout illustration.

3.3.3 Alphabetical index. (See figure 2) An alphabetical index shall be prepared and placed in the front of the manual following the "List of Effective Pages" page (s). An existing alphabetical index in the rear portion of the manual shall be deleted. Subjects contained in the existing alphabetical index may be used in constructing a new alphabetical index. The listing shall be single column. The first page of the index shall contain the title "ALPHABETICAL INDEX" centered at the top of the page. Each page of the alphabetical index shall include a "subject" heading, a "paragraph/figure/table number" heading, a "page number" heading, and a "search number" heading. Main entries in the "subject" column shall begin at the left margin. Subordinate entries listed below a main entry shall be indented two spaces from the left margin.

3.3.3.1 Alphabetical index content. The alphabetical index shall be so prepared as to enable the user to easily locate an item, figure, table, information, task, or function contained in the text of the manual. Subjects shall be arranged with the noun name preceding the modifiers; for example, "Engine Driven Fuel Pump" should be listed as "Pump, Fuel, Engine Driven". Titles may also be listed under significant modifiers as cross reference entries; for example, "Fuel Pump, Engine Driven". The pump should also be entered as a subordinate entry under the "Engine Fuel System". Subsystem and equipment components shall be listed alphabetically as subordinate entries to the system or equipment of which they are a part and also as main entries in the overall alphabetical listing. The task or function such as assembly, cleaning, disassembly, inspection, repair, etc. shall be listed as subordinate listings to the main entries, as applicable.

3.3.3.2 Alphabetical index page numbers. Page numbers for the alphabetical index shall be consecutively numbered in Arabic Numerals with the word "Index" preceding the page number.

3.3.3.3 Search number. A search number identifies the microfilm frame location of information in the technical manual from the first page (zero page) of the alphabetical index. Search numbers shall be inserted in the alphabetical index. The first page of the alphabetical index shall contain the following instructions for use of search numbers at the bottom of the page:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

SEARCH INSTRUCTIONS

Auto Reader - Printer: Push "0", push "RESET", enter Search Number, push "SRCH".
Portable Reader: Add Search number to frame number of this page and advance.

The remaining pages of the alphabetical index shall contain the following instructions at the bottom of each page:

SEARCH INSTRUCTIONS

Auto Reader - Printer: Return to page 1 of this index, push "0", push "RESET", enter Search number, push "SRCH".
Portable Reader: Add Search Number to Frame Number of page 1 of this index and advance.

3.3.4 Numerical index cross reference. (See figure 3) All IPB's containing more than twenty Numerical Index pages shall include a Numerical Index Cross Reference (NICR). The NICR shall list the first part number that appears in the "Numerical Index" and the first number that appears on every second page of the "Numerical Index". The NICR shall include "part number", "page number", and "search number" headings in two column format. The NICR shall be placed immediately before the "Numerical Index". Page numbers shall be assigned to the NICR in such a manner so as to indicate their relationship to the Numerical Index; for example, if the Numerical Index is Section 3 of an IPB Manual, the NICR pages shall be numbered 3-00, 3-01, 3-02, etc. The first page of the NICR shall be considered a zero page and include search instructions in accordance with figure 3. The remaining pages of the NICR shall include search instructions contained in figure 1-continued. The search numbers entered in the NICR shall represent the number of frames required to advance the microfilm from the first page of the NICR to the page the part number appears in the Numerical Index.

3.3.5 Reference designation index cross reference. (See figure 4) All IPB's containing more than twenty pages of reference designations shall include a Reference Designation Index Cross Reference (RDICR). The RDICR shall list the first reference designation that appears in the Reference Designation Index and the first reference designation that appears on every second page of the Reference Designation Index. The RDICR shall include "reference designation", "page number", and "search number" headings in two column format. The RDICR shall be placed immediately before the Reference Designation Index. Page numbers shall be assigned to the RDICR in such a manner so as to indicate their relationship to the Reference Designation Index; for example, if the Reference Designation Index is Section 4 of an IPB Manual, the RDICR pages shall be numbered 4-00, 4-01, etc. The first page of the RDICR shall be considered a zero page and include search instructions at the bottom of the page in accordance with figure 4. The remaining pages of the RDICR shall include search instruction contained in figure 1-continued. The search number entered in the RDICR shall represent the number of frames required to advance the microfilm from the first page of the RDICR to the page the reference designation appears in the Reference Designation Index.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Contractor inspection. Material furnished in accordance with this specification shall be inspected by the contractor for conformance to the applicable requirements herein.

5. PACKAGING

5.1 Preparation for delivery. Packaging, packing, and marking for shipment shall be in accordance with MIL-M-38784.

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. The technical manuals are intended for microfilming and use under the Maintenance Information Automated Retrieval System (MIARS).

6.2 Ordering Data.

6.2.1 Procurement requirements. Procurement documents should speify.

a. Title, number, and date of this specification.
b. Responsibility for inspection if other than specified in 4.1.

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of Defense Acquisition Regulations (DAR), the data requirements identified below will be developed as specified by an approved data item description (DD Form 1664) and delivered in accordance with the approved contract data requirements list (DD Form 1423) incorporated in the contract, when the provisions of DAR are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data	Applicable DID
3.2	Microfilm compatible technical manual	UDI-M-21368

(Copies of the DID required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.3 Definitions.

6.3.1 Frame. A frame is the total area of microfilm utilized in one exposure, regardless of whether or not the area is filled by the document image area.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

6.3.2 Search number. A search number is the number of frames required to advance from the first page of an index to the page containing the desired data.

6.3.3 Zero page. A page identified as a starting point to determine the location of information contained in the manual or a portion of the manual. The first page of the alphabetical index is a Zero page.

6.3.4 Requiring activity. The organization of a using military service or the organization delegated by a using service, which is responsible for the selection of and determination of requirements for a specific support element.

6.4 Specification figures. The figures provided in this specification are intended to illustrate methods of technical data presentation. They shall not be interpreted as limiting the technical content requirements that are established by the text matter to which the figures apply.

Preparing activity:

Navy - AS

(Project TMSS-N149)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

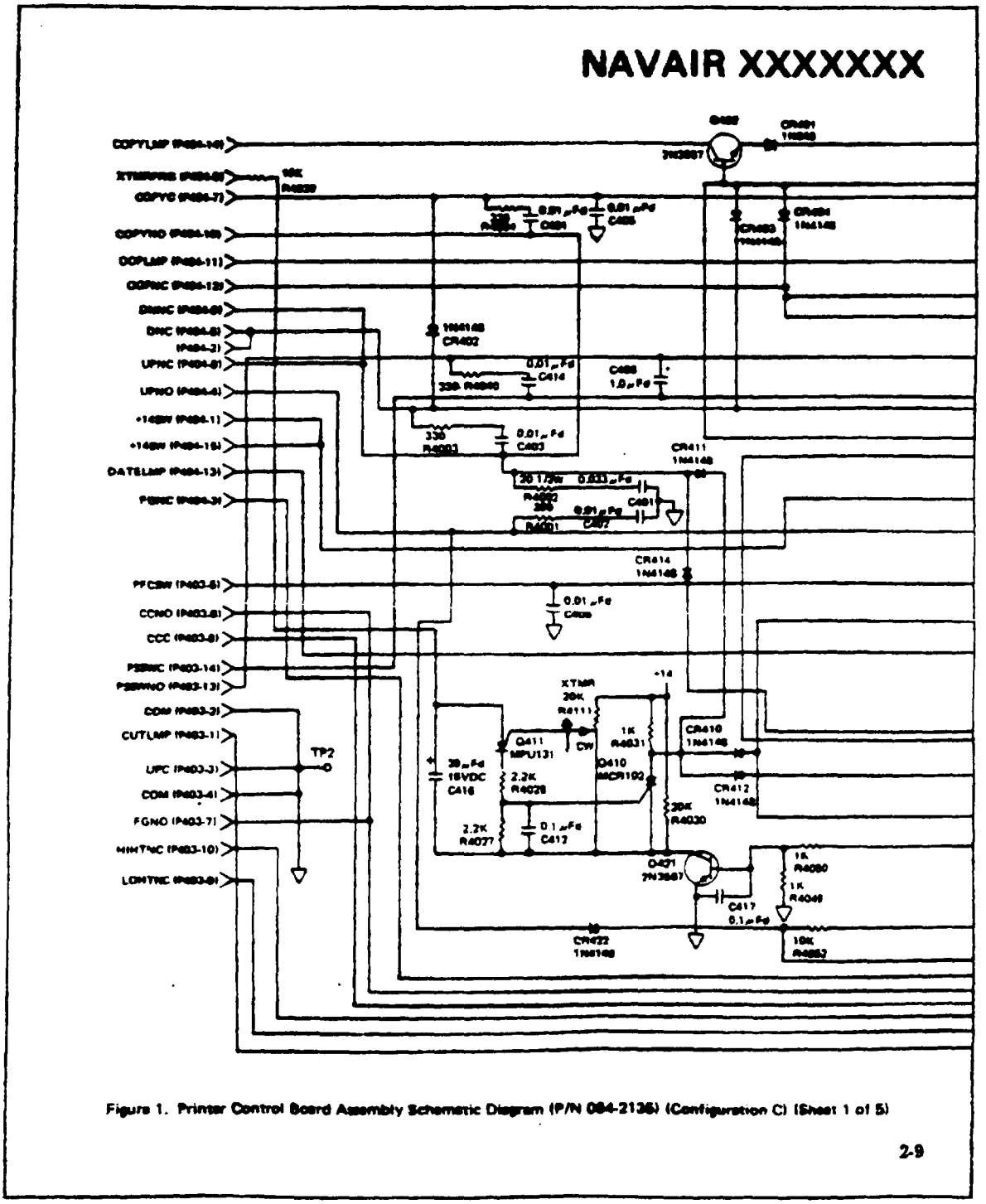


Figure 1. Printer Control Board Assembly Schematic Diagram (P/N 084-2136) (Configuration C) (Sheet 1 of 5)

2-9

FIGURE 1. Example of a foldout illustration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

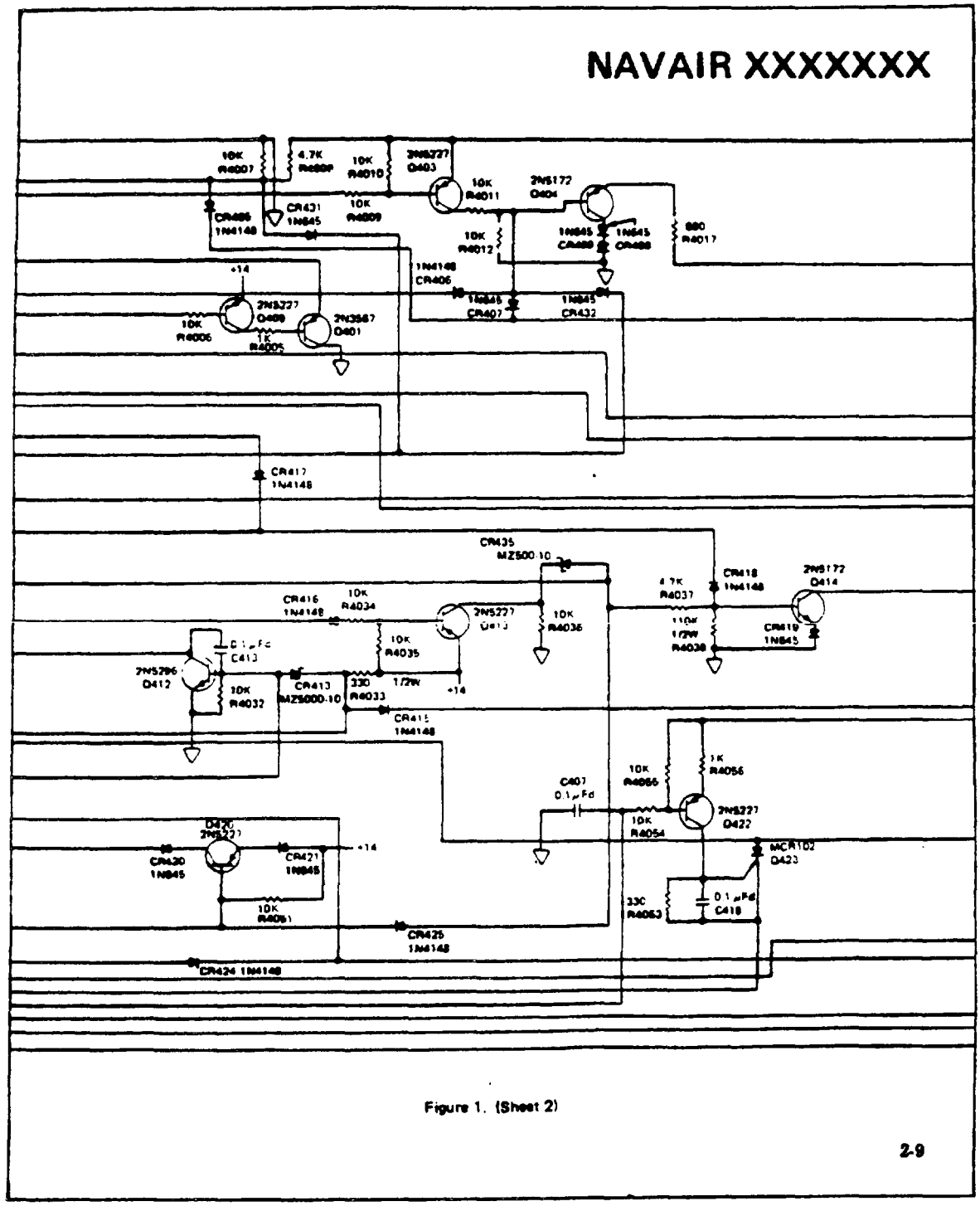


FIGURE 1. Example of a foldout illustration—Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

NAVAIR XXXXXXXX

ALPHABETICAL INDEX

Subject	Paragraph, Figure Table Number	Page Number	Search Number
Accumulator, Combined			
Checkout	3-153	5-51	251
Installation	3-155	5-52	252
Removal	3-154	5-51	251
Accumulator, Flight			
Checkout	4-24	4-8	134
Installation	4-26	4-10	136
Removal	4-25	4-10	136
Actuator, Radome Fold			
Installation	6-61	6-21	297
Removal	6-60	6-20	296
Bleed Valve, Relief			
Installation	4-187	4-62	188
Removal	4-186	4-62	188
Check Valves, Combined			
Installation	5-76	5-26	227
Removal	5-75	5-25	226
Combined Hydraulic System			
Check Valves	5-75	5-25	226
Cooler, Hydraulic Oil	4-213	4-71	196
Pump, Engine Driven	5-6	5-2	202
Reservoir, Hydraulic	4-174	4-58	184
Switch, Pressure	5-108	5-36	236
Cooler, Hydraulic Oil			
Installation	4-215	4-72	197
Removal	4-214	4-71	196
Filter Elements			
Installation	4-107	4-36	162
Inspection	4-106	4-36	161
Removal	4-105	4-35	161
Flow Regulator, Flight			
Installation	4-55	4-18	144
Removal	4-54	4-18	144

SEARCH INSTRUCTIONS

Auto Reader-Printer: Push "0," push "RESET," enter Search Number, push "SRCH."

Portable Reader: Add Search Number to Frame Number of this page and advance.

FIGURE 2. Example of an alphabetical index.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

NAVAIR XXXXXX

ALPHABETICAL INDEX

Subject	Paragraph, Figure Table Number	Page Number	Search Number
Gearbox Assembly			
Assembly	6-13	6-8	353
Cleaning	6-10	6-3	348
Disassembly	6-9	6-3	348
Inspection	6-11	6-4	349
Installation	6-15	6-12	357
Removal	6-8	6-3	348
Repair	6-12	6-5	350
Test	6-14	6-10	355
Housing, Bearing			
Cleaning	5-66	5-23	255
Inspection	5-67	5-23	255
Installation	5-68	5-24	256
Removal	5-65	5-22	254
Introduction	1-1	1-1	4
Oil Pump			
Assembly	4-35	4-17	167
Cleaning	4-32	4-13	163
Disassembly	4-31	4-12	162
Inspection	4-33	4-13	163
Installation	4-37	4-21	171
Removal	4-30	4-12	162
Repair	4-34	4-14	164
Test	4-36	4-19	169

SEARCH INSTRUCTIONS

Auto Reader-Printer: Return to page 1 of this index, push "0," push "RESET," enter Search Number, Push "SRCH."

Portable Reader: Add Search Number to Frame Number of page 1 of this index and advance.

FIGURE 2. Example of an alphabetical index—Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

NAVAIR XXXXXX

NUMERICAL INDEX CROSS REFERENCE

Part No.	Page No.	Search No.	Part No.	Page No.	Search No.
AA	3-2	6	3230019	3-58	62
AN392	3-4	8	3231012-67	3-60	64
AN632-4N	3-6	10	3231043	3-62	66
AN951-2A	3-8	12	32311517	3-64	68
AN965N5	3-10	14	3231470	3-66	70
G587634	3-12	15	3235987	3-77	72
313983	3-40	44	3238764	3-90	90
314680	3-42	46	3280235	3-98	102
3211044	3-44	48	3285001	3-100	104
3211250	3-46	50	3289016	3-102	106
3211506	3-48	52	3289115	3-104	108
3215155-03	3-50	54	3294524-3	3-106	110
3218011	3-52	56	3296516	3-108	112
3219477	3-54	58	3298940	3-110	114
3223005	3-56	60	330066	3-112	116

SEARCH INSTRUCTIONS
Auto-Reader-Printer: Push "0," push "RESET," enter Search Number, push "SRCH."
Portable Reader: Add Search Number to Frame Number of this page and advance.

3-00

FIGURE 3. Example of a numerical index cross reference.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

NAVAIR XXXXXXX

REFERENCE DESIGNATION INDEX CROSS REFERENCE

Reference Designation	Page No.	Search No.	Reference Designation	Page No.	Search No.
BX1801	4-1	1	R1379	4-21	21
CR311	4-3	3	R2204	4-23	23
C1404	4-5	5	R314	4-25	25
C1635	4-7	7	R520	4-27	27
C2415	4-9	9	R741	4-29	29
C604	4-11	11	TP305	4-31	31
FL1025	4-13	13	YDS2001	4-33	33
K1001	4-15	15	Y1212	4-35	35
L1635	4-17	17	Y1310	4-37	37
RT301	4-19	19	Y1522	4-39	39

SEARCH INSTRUCTIONS

Auto-Reader-Printer: Push "0," push "RESET," enter Search Number, push "SRCH."

Portable Reader: Add Search Number to Frame Number of this page and advance.

4-00

FIGURE 4. Example of a reference designation index cross reference.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

ALPHABETICAL INDEX

A

Alphabetical index 3.3.3
Alphabetical index content 3.3.3.1
Alphabetical index page numbers 3.3.3.2
Applicable documents 2
Artwork 3.3.2

C

Classification markings - foldout illustrations 3.3.2.7.4
Color 3.3.2.4
Contractor inspection 4.2

D

Darkness and sharpness of lines 3.3.2.2.1
Data 3.2
Data requirements 6.2.2
Definitions 6.3

E

Engineering drawings 3.3.2.6

F

Foldout-foldup illustrations 3.3.2.7.1
Foldout illustrations 3.3.2.7
Foldout illustrations increment markings 3.3.2.7.3
Frame 6.3.1

G

General 3.1

I

Intended use 6.1
Issue of documents 2.1

L

Line drawings 3.3.2.2
Location of foldout illustrations 3.3.2.7.2

M

Microfilm compatibility criteria 3.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

N

Notes 6
Numerical index cross reference 3.3.4

O

Ordering data 6.2
Other publications 2.2
Overlays 3.3.2.5

P

Packaging 5
Photographs 3.3.2.1
Preparation for delivery 5.1
Procurement requirements 6.2.1

Q

Quality assurance provisions 4

R

Reference designation index cross reference 3.3.5
Requirements 3
Responsibility for inspection 4.1
Requiring activity 6.3.4

S

Scope 1,1.1
Search number 6.3.2
Search numbers 3.3.3.3
Sidewise illustrations 3.3.2.3
Specification figures 6.4

T

Type size 3.3.1

Z

Zero page 6.3.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Department (ESSD)
Code 93
Lakehurst, NJ 08733

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-M-85401(AS) Manuals, Technical, Conventional to Microfilm Compatible, Conversion of

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:46Z
Check the source to verify that this is the current version before use.